

Critical Time Thresholds in Online Chess Decision-Making: A Lichess-Based Study of Move Accuracy Under Time Pressure

Abstract

[Insert final abstract after results are complete.]

1. Introduction

Chess is often described as a game of calculation, pattern recognition, and strategic planning, but competitive chess is also a game of time management. In online chess, players frequently make decisions with only seconds remaining on the clock. Coaches and players often assume that “time trouble” leads to worse moves, yet the size and shape of this effect are less clear. Does accuracy decrease gradually as time runs low, or is there a critical threshold below which mistakes become much more likely?

This paper studies the relationship between remaining clock time and move accuracy in online chess. Using publicly available Lichess data, I analyze blitz and rapid games and measure move quality using engine-based centipawn loss and blunder probability. The goal is to test whether low remaining time predicts inaccurate moves, and whether this relationship differs across time controls.

2. Research Question and Hypotheses

The main research question is:

Does low remaining clock time increase the probability of inaccurate moves in online blitz and rapid chess games?

I test three hypotheses:

H1: Moves made with less remaining time have higher average centipawn loss.

H2: Blunder probability increases as remaining clock time decreases.

H3: The effect of low remaining time is nonlinear, with a sharper increase in blunder probability below critical time thresholds.

3. Data

The study uses publicly available games from Lichess. The dataset includes rated blitz and rapid games with clock information. Games are filtered to remove bots, variants, extremely short games, and games without usable clock data.

Each move-level observation includes:

- game ID
- player color
- player rating
- player title, if available
- time control
- move number
- remaining clock time
- engine evaluation before and after the move
- centipawn loss
- blunder indicator

4. Methods

For each move, move accuracy is measured using centipawn loss, defined as the decrease in engine evaluation caused by the played move relative to the best available evaluation. A move is classified as a blunder if its centipawn loss exceeds a predefined threshold, such as 200 centipawns.

Remaining time is grouped into bins, such as:

- more than 60 seconds
- 30–60 seconds
- 10–30 seconds
- 5–10 seconds
- less than 5 seconds

The analysis compares mean and median centipawn loss across time bins and estimates blunder probability under different remaining-time conditions. Regression models are used to test whether remaining time predicts blunders while controlling for rating, move number, and time control.

5. Results

[To be completed after data analysis.]

Expected figures:

Figure 1. Distribution of remaining time by time control.

Figure 2. Median centipawn loss by remaining-time bin.

Figure 3. Blunder probability by remaining-time bin.

Figure 4. Comparison of blitz and rapid curves.

Table 1. Regression model predicting blunder probability.

6. Discussion

[To be completed after results.]

This section will interpret whether the data supports the hypothesis that time pressure reduces move accuracy, whether the effect appears linear or threshold-based, and whether the effect differs between blitz and rapid games. It will also discuss possible explanations, including cognitive load, premove behavior, time-control adaptation, and position complexity.

7. Limitations

Several limitations should be considered. First, online chess differs from over-the-board chess, so results may not generalize to all competitive settings. Second, engine-based centipawn loss is an imperfect measure of human move quality, especially in winning or losing positions. Third, remaining time may be correlated with position difficulty: players may spend more time in complicated positions, which can make causality difficult to establish. Finally, Lichess players may not represent all chess players.

8. Conclusion

[To be completed after results.]

References

[To be completed.]

Research Decisions

Dataset:

- Platform: Lichess
- Game types: blitz and rapid
- Game status: rated games only
- Players: titled or high-rated players
- Excluded: bots, variants, games without clock data, very short games

Unit of analysis:

- One row = one move

Main variables:

- remaining clock time
- time control
- move number
- player rating
- engine evaluation before and after move
- centipawn loss
- blunder flag

Accuracy metric:

- centipawn loss
- blunder = centipawn loss above threshold, initially 200 centipawns

Time bins:

- >60 seconds
- 30–60 seconds
- 10–30 seconds
- 5–10 seconds
- <5 seconds

Main hypothesis:

- Blunder probability increases when remaining clock time becomes low, especially below critical thresholds.

